Do Discounts Influence Consumer Spending Behavior?

In this project, I aim to analyze how consumer spending behavior changes during large promotional events such as Black Friday. I will use a publicly available Black Friday dataset that includes customer information such as age, gender, occupation, and city category, along with product categories and purchase amounts. The dataset contains a large number of observations, which makes it suitable for statistical analysis and machine learning applications.

Since the main dataset does not directly include discount information, I will use product categories as proxy variables to examine the possible effects of discounts on consumer spending behavior. Different product categories may reflect different pricing and promotional strategies, which can provide indirect insights into discount-related effects. To strengthen the analysis, I will also incorporate an additional e-commerce pricing dataset containing category-level price and discount information as a supplementary data source.

The project will include data cleaning, exploratory data analysis (EDA), and hypothesis testing to examine the relationships between demographic variables, product categories, and purchase amounts. In particular, I will apply statistical methods such as t-tests and ANOVA to identify whether there are significant differences in consumer spending across different groups. In addition, I will apply a machine learning regression model to predict purchase amounts and evaluate which features are the strongest predictors of spending behavior.

This topic is interesting to me because I have previously done a presentation on a similar subject in one of my courses. During that presentation, I became interested in how marketing strategies, especially discounts, influence purchasing decisions. Therefore, I would like to explore this topic further using real-world data and data science techniques to better understand consumer behavior.